

# Hengrong Du

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| CONTACT INFORMATION | Fisk University<br>Department of Math & Computer Science                                                                                                        | Page: <a href="https://hengrongdu.netlify.app/">https://hengrongdu.netlify.app/</a><br>E-mail: <a href="mailto:hdu@fisk.edu">hdu@fisk.edu</a> |
| RESEARCH INTERESTS  | Partial Differential Equations for Machine Learning and Complex Fluids                                                                                          |                                                                                                                                               |
| ACADEMIC POSITIONS  | <b>Fisk University</b> , Nashville, TN, USA<br>Assistant Professor, <a href="#">Department of Mathematics and Computer Science</a>                              | August 2025-                                                                                                                                  |
|                     | <b>University of California, Irvine</b> , Irvine, CA, USA<br>Visiting Assistant Professor, <a href="#">Department of Mathematics</a>                            | July 2024-July 2025                                                                                                                           |
|                     | <b>Vanderbilt University</b> , Nashville, TN, USA<br>Postdoc Scholar (Research), <a href="#">Department of Mathematics</a><br>Mentor: <b>Dr. Gieri Simonett</b> | Sep. 2021-June 2024                                                                                                                           |
| EDUCATION           | <b>Purdue University</b> , West Lafayette, IN<br>Ph.D., Mathematics<br>Advisor: <b>Dr. Changyou Wang</b>                                                        | Aug. 2016-Aug. 2021                                                                                                                           |
|                     | <b>Beijing Normal University</b> , Beijing, China<br>M.S., Computational Mathematics<br>Advisor: <b>Dr. Jiequan Li</b>                                          | Sep. 2013-July 2016                                                                                                                           |
|                     | <b>South China Normal University</b> , Guangzhou, China<br>B.S., Mathematics and Applied Mathematics                                                            | Sep. 2009-July 2013                                                                                                                           |

## PUBLICATIONS

### Diffusion and Optimal Transport in Machine Learning

1. Xinyang Liu, Hengrong Du, Wei Deng, Ruqi Zhang. *Optimal Stochastic Trace Estimation in Generative Modeling* . AISTATS 2025.
2. Yikun Bai, Rocio Diaz Martin, Hengrong Du, Ashkan Shahbazi, Soheil Kolouri. *Linear Partial Gromov-Wasserstein Embedding* . ICLR 2025.
3. Yikun Bai, Rocio Diaz Martin, Hengrong Du, Ashkan Shahbazi, Soheil Kolouri. *Partial Gromov-Wasserstein Metric* . ICLR 2025.
4. Haoyang Zheng, Hengrong Du, Qi Feng, Wei Deng, Guang Lin. *Constrained exploration via reflected replica exchange stochastic gradient Langevin dynamics*. ICML 2024.
5. Wei Deng, Yu Chen, Nicole Tianjiao Yang, Hengrong Du, Qi Feng, Ricky T. Q. Chen. *Reflected Schrödinger bridge for constrained generative modeling*. UAI 2024.

6. Wei Deng, Yu Chen, Nicole Tianjiao Yang, Hengrong Du, Qi Feng, Ricky T. Q. Chen. *On convergence of approximate Schrödinger bridge with bounded cost*. ICML Workshop on New Frontiers in Learning, Control, and Dynamical Systems (2023).

### Complex Fluids

1. Hengrong Du, Gieri Simonett, Yuanzhen Shao. *On a thermodynamically consistent model for magnetoviscoelastic fluids in 3D*. J. Evol. Equ. 24 (2024), no. 9, 1-51.
2. Hengrong Du, Yuanzhen Shao, Gieri Simonett. *Well-posedness for magneto-viscoelastic fluids in 3D*. Nonlinear Anal. Real World Appl. 69 (2023), no. 103759.
3. Hengrong Du, Tao Huang, Changyou Wang. *Weak compactness property of simplified nematic liquid crystal flows in dimension two*. Math. Z. 302 (2022), no. 2, 2111-2130.
4. Hengrong Du, Changyou Wang. *Global weak solutions to the stochastic Ericksen–Leslie system in dimension two*. Discrete Contin. Dyn. Syst. 42 (2022), no. 5, 2175-2197.
5. Hengrong Du, Changyou Wang. *Partial regularity of a nematic liquid crystal model with kinematic transport effects*. Nonlinearity 34 (2021), no. 5, 3001-3045.
6. Hengrong Du, Yimei Li, Changyou Wang. *Weak solutions of non-isothermal nematic liquid crystal flow in dimension three*. J. Elliptic Parabol. Equ. 6 (2020), no. 1, 71-98.
7. Hengrong Du, Xianpeng Hu, Changyou Wang. *Suitable Weak Solutions for the Co-rotational Beris–Edwards System in Dimension Three*. Arch. Ration. Mech. Anal. 238 (2020), no. 2, 749-803.

### Calculus of Variations

1. Hengrong Du, Nung Kwan Yip. *Stability of self-similar solutions to geometric flows*. Interfaces Free Bound. 25 (2023), no. 2, 155–191.
2. Hengrong Du, Qinfeng Li, Changyou Wang. *Compactness of  $M$ -uniform domains and optimal thermal insulation problems*. Adv. Calc. Var. 16 (2023), no. 1, 17-43.

## PREPRINTS

1. Hengrong Du, Qi Feng, Changwei Tu, Xiaoyu Wang, Lingjiong Zhu. *Non-Reversible Langevin Algorithms for Constrained Sampling*. [arXiv:2501.11743](#).
2. Yikun Bai, Huy Tran, Hengrong Du, Xinran Liu, Soheil Kolouri. *Fused Partial Gromov-Wasserstein for Structured Objects*. [arXiv:2401.03662](#).
3. Hengrong Du, Chuntian Wang. *Partial regularity for the three-dimensional stochastic Ericksen–Leslie Equations*. [arXiv:2401.03662](#).

## AWARDS

- Bilsland Dissertation Fellowship June 2020-May 2021  
It was intended to give the most accomplished final-year PhD candidates an opportunity to complete the dissertation within the 2020–21 academic year by devoting full-time effort to research and writing.

## TALKS

- PDE & Applied Mathematics seminar  
University of California, Riverside January 2025
- PDEs & Applications Seminar  
Queen's University January 2025
- Applied Math Seminar  
The University of Alabama October 2024
- PDEs and Fluid Mechanics and Atmospheric Sciences  
West Virginia University April 2024
- AMS Special Session on Fluids: Analysis, Applications, and Beyond  
Florida State University March 2024
- Financial Mathematics Seminars  
Florida State University March 2024
- AMS Special Section on Dynamics and Regularity of PDEs  
Joint Mathematics Meeting January 2024
- Seminar on Analysis and Stochastic Analysis  
Auburn University November 2023
- PDE Seminar  
University of Tennessee, Knoxville October 2023
- PDE/Applied Math Seminar  
Indiana University, Bloomington October 2023
- Richard F. Barry Jr. Seminar Series  
Old Dominion University September 2023
- The 13th AIMS Conference  
University of North Carolina Wilmington June 2023
- AMS Spring Central Sectional Meeting  
University of Cincinnati April 2023
- Undergraduate Seminar in Mathematics  
Vanderbilt University March 2023
- Analysis Seminar  
Wayne State University March 2023

- 3rd Biennial Meeting of SIAM Pacific Northwest Section  
Washington State University May 2022
- AMS Spring Central Virtual Sectional Meeting  
Purdue University March 2022
- Graduate Online Mini-course  
Beijing Normal University November 2021
- PDE Seminar Fall 2021  
Vanderbilt University September 2021
- Analysis Seminar  
The University of Alabama February 2021
- PDE Seminar  
Purdue University September 2020
- Student Analysis Seminar  
Purdue University February 2020
- Student Analysis Seminar  
Purdue University February 2020
- PDE Seminar  
Purdue University January 2020

CONFERENCE  
PARTICIPATION

- Lean for PDEs, an ICARM & SLMATH collaboration (Virtual)  
The Simons Laufer Mathematical Sciences Institute Oct. 6-9, 2025
- Summer School on Theory and Applications of Elliptic PDE  
University of California, Irvine June 23-27, 2025
- Rivière-Fabes Symposium on Analysis and PDE  
University of Minnesota Apr. 19-21, 2023
- Shanks Workshop on Advances in Mathematical and Theoretical Biology  
Vanderbilt University Mar. 17-19, 2023
- Workshop on Geometry and Analysis of Fluid Flows  
Stony Brook University Jan. 16-20, 2023
- AMS Fall Central Sectional Meeting  
University of Texas at El Paso Sep. 17-18, 2022
- Shanks Workshop on Mathematical Aspects of Fluid Dynamics  
Vanderbilt University Feb. 19-20, 2022
- KUMUNU-ISU Conference on PDE, Dynamical Systems, and Applications  
University of Nebraska-Lincoln Oct. 23-24, 2021
- AMS Fall Western Sectional Meeting  
Online Oct. 23-24, 2021
- The 84th Midwest PDE Seminar  
Illinois Institute of Technology Oct. 26-27, 2019
- Midwest Geometry Conference 2019  
Iowa State University Sept. 6-8, 2019
- The 83rd Midwest PDE Seminar  
Indiana University Bloomington Mar. 30-31, 2019

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|------------------------|-----------------------------------------------------------|-----------------------|
| TEACHING<br>EXPERIENCE | Instructor at Fisk University                             | Fall 2025-Present     |
|                        | • MATH 125 Discrete Mathematics                           | Fall 2025             |
|                        | • MATH 130 Calculus II                                    | Fall 2025             |
|                        | Instructor at University of California, Irvine            | Fall 2024-Spring 2025 |
|                        | • MATH 3A Introduction to Linear Algebra                  | Spring 2025           |
|                        | • MATH 112B Introduction to Partial Differential Equation | Winter 2025           |
|                        | • MATH 3D Elementary Differential Equations               | Fall 2024             |
|                        | Instructor at Vanderbilt University                       | Fall 2021-Present     |
|                        | • MATH 3100 Introduction to Analysis                      | Spring 2024           |
|                        | • MATH 2400 Differential Equations with Linear Algebra    | Fall 2023             |
|                        | • MATH 3100 Introduction to Analysis                      | Spring 2023           |
|                        | • MATH 3100 Introduction to Analysis                      | Fall 2022             |
|                        | • MATH 1301 Accelerated Single-Variable Calculus II       | Fall 2022             |
|                        | • MATH 2420 Methods of Ordinary Differential Equations    | Spring 2022           |
|                        | • MATH 2400 Differential Equations with Linear Algebra    | Spring 2022           |
|                        | Recitation Instructor at Purdue University                | Fall 2016-Summer 2021 |
|                        | • MA 26200 Linear Algebra And Differential Equations      | Fall 2019             |
|                        | • MA 16200 Plane Analytic Geometry And Calculus II        | Spring 2019           |
|                        | • MA 16500 Plane Analytic Geometry And Calculus I         | Fall 2018             |